

Experiment No. 6: Predictive Analytics using Linear Regression and Model Performance Evaluation

Aim

To develop a Linear Regression model using Python for predicting continuous outcomes and evaluate its performance using appropriate regression metrics.

Questions & Answers

Q1. What is Predictive Analytics, and how is it used in real-world applications?

Answer:

- **Predictive Analytics:** The branch of advanced analytics that utilizes historical data, statistical algorithms, and machine learning techniques to identify the likelihood of future outcomes.
- **Real-World Applications:**
 - **Finance:** Credit scoring, fraud detection, and stock price forecasting.
 - **Healthcare:** Predicting patient readmission rates and disease outbreaks.
 - **Marketing:** Customer churn prediction and targeted advertising.

Q2. Explain the working principle of the Linear Regression algorithm.

Answer:

Linear Regression is a supervised learning algorithm used to model the linear relationship between independent variables (\$X\$) and a continuous target variable (\$y\$).

- **Mathematical Equation:**

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

Where:

- y = Target variable
- β_0 = y -intercept
- $\beta_1, \beta_2, \dots, \beta_n$ = Regression coefficients
- ϵ = Error term
- **Optimization:** The algorithm fits a straight line (or hyperplane) by minimizing the **Sum of Squared Errors (SSE)** between the actual values and predicted values using methods like **Ordinary Least Squares (OLS)** or **Gradient Descent**.

Q3. Differentiate between dependent and independent variables with suitable examples.

Answer:

Feature	Independent Variable (\$X\$)	Dependent Variable (\$y\$)
Definition	Inputs/predictors manipulated or measured to predict the output.	Target/outcome metric being predicted.
Role	Serves as the cause or input factor.	Serves as the effect or response variable.

Feature	Independent Variable (\$X\$)	Dependent Variable (\$y\$)
Example (Housing)	House Area (SqFt), Number of Bedrooms, Age of House.	House Sale Price.
Example (Sales)	Advertising Budget spent on TV/Social Media.	Total Units Sold.

Q4. Why is it necessary to split the dataset into training and testing sets?

Answer:

- **Generalization Assessment:** Evaluates how accurately the model performs on unseen, real-world data.
- **Overfitting Prevention:** Ensures the algorithm is not merely memorizing training samples, but learning real underlying patterns.
- **Model Validation:** Provides an unbiased estimate of performance metrics (\$MAE\$, \$RMSE\$, \$R^2\$) prior to production deployment.

Q5. What is the significance of the \$R^2\$ Score in regression analysis?

Answer:

The **\$R^2\$ Score (Coefficient of Determination)** measures the proportion of total variance in the dependent variable that is predictable from the independent variables.

- **Formula:**
$$R^2 = 1 - \frac{\text{Sum of Squared Residuals (SSR)}}{\text{Total Sum of Squares (SST)}}$$
- **Interpretation:**
 - Range: \$0\$ to \$1\$ (or \$0\%\$ to \$100\%\$).
 - An \$R^2\$ of **0.85** implies that \$85\%\$ of the variance in the target variable (e.g., House Price) is explained by the model's feature set.

Q6. Differentiate between MAE, MSE, and RMSE. Which metric is more sensitive to large prediction errors?

Answer:

- **Mean Absolute Error (MAE):** Average absolute difference between actual and predicted values. It treats all error sizes linearly.
$$\text{MAE} = \frac{1}{n} \sum |y - \hat{y}|$$
- **Mean Squared Error (MSE):** Average of squared errors. Squaring removes negative signs but alters measurement units to squared values.
$$\text{MSE} = \frac{1}{n} \sum (y - \hat{y})^2$$
- **Root Mean Squared Error (RMSE):** Square root of MSE. Converts units back to original target scale.
$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (y - \hat{y})^2}$$
- **Sensitivity:** **MSE and RMSE** are significantly more sensitive to large prediction errors/outliers because the residuals are squared prior to averaging.

Q7. What assumptions should be satisfied before applying Linear Regression?

Answer:

1. **Linearity:** Linear relationship between features and target.

2. **Independence:** Observations/errors are independent (no autocorrelation).
3. **Homoscedasticity:** Constant variance of residuals across all predictor levels.
4. **Normality:** Residuals follow a normal distribution.
5. **No Multicollinearity:** Predictor variables are not highly correlated with each other.

Q8. How can overfitting and underfitting affect the performance of a regression model?

Answer:

- **Underfitting (High Bias):**
 - Model is too simple to capture patterns.
 - *Effect:* Poor performance on both training and test data.
- **Overfitting (High Variance):**
 - Model learns noise and random fluctuations in training data.
 - *Effect:* Exceptionally low training error, but high error on unseen test data.

Q9. Mention any three real-world applications of Linear Regression in business or industry.

Answer:

1. **Real Estate:** Estimating property valuation based on location, square footage, and amenities.
2. **Retail & Sales:** Forecasting seasonal demand and sales revenue based on marketing expenditure.
3. **Insurance:** Setting premium prices based on age, medical history, and risk factors.

Q10. How can feature selection improve the accuracy and interpretability of a predictive model?

Answer:

- **Reduces Multicollinearity:** Eliminates redundant features that inflate standard errors of regression coefficients.
- **Mitigates Overfitting:** Simplifies model complexity, improving generalization on test data.
- **Speeds Up Training:** Reduces computational overhead and data storage requirements.
- **Enhances Interpretability:** Focuses business insights on key drivers influencing the target metric.